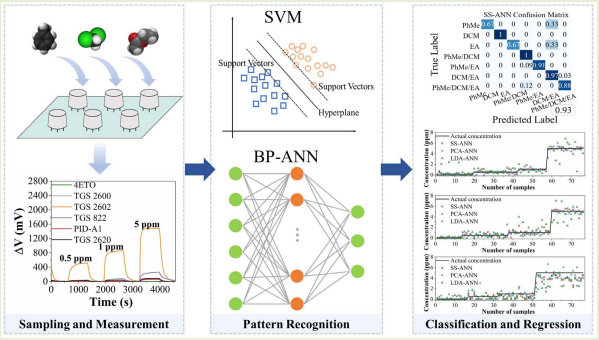
Hybrid Gas Sensor Array to Identify and Quantify Low-Concentration VOCs Mixtures Commonly Found in Chemical Industrial Parks

Linjie Xu, Longchao Yao, *Member, IEEE*, Yongguang Wang, Weihong Wu, Wenhao Lin, Chenghang Zheng, Yuanqun Feng, and Xiang Gao

***Abstract*—The chemical industrial park is an important source of volatile organic compounds (VOCs) emissions which is difficult to monitor due to its diversity and low concentration. In this work, a hybrid sensor array consisting of six gas sensors (four semiconductor sensors, one electrochemical sensor, and one photo ionization detector) was constructed to measure VOCs. Three most frequently detected VOCs in a Chinese fine chemical industrial park, namely toluene, dichloromethane, ethyl acetate, and their mixtures, were tested all below 5 ppm (even below 1 ppm in many cases). Support vector machine (SVM) and back-propagation artificial neural network (BP-ANN) based on different input features, including steady-state responses (SS), the first four components preprocessed by principal component analysis (PCA) and linear discriminant analysis (LDA), were trained and evaluated. The 10-fold cross validation results indicate that SS-SVM and SS-ANN are two best models with accuracy and R^2 of about 94% and 0.95 for the validation set, respectively. Compared with LDA, PCA is more suitable for dimension reduction in this study, since the performance of models with the first 4 PCs are closer to that of models with SS. The results demonstrate that the proposed hybrid sensor array system with SVM and ANN models is able to identify and quantify low-concentration VOCs mixtures. This method is promising to be used for low-cost and online monitoring of VOCs emissions in chemical industrial parks.**



***Index Terms*—Low-concentration VOCs mixture, component identification, concentration estimation, hybrid sensor array.**

I. Introduction

VOLATILE organic compounds (VOCs), such as alkanes, ketones, esters, aldehydes, aromatic hydrocarbons, etc., are important precursors of secondary organic aerosols (SOA) and surface ozone. Moreover, various kinds of VOCs have pungent smell and are toxic and harmful to human beings [1]–[3]. There are diverse sources of VOCs emissions, including industrial sources, agricultural sources, domestic sources, and so on, among which the industrial source is crucial to air pollution. Large numbers of chemical enterprises are concentrated in chemical industrial parks, including pharmaceutical manufacturing, dye and auxiliary production, pesticide production, etc. In these industries, large numbers of raw materials, solvents, and products containing VOCs are produced, transported, stored, and used. Hence, chemical industrial park is one of the main sources of VOCs emissions. Measurement and monitoring of VOCs have caused more and more attention due to the stringent environmental regulations and increasing health concerns. However, vast numbers of VOCs that can be found in chemical industrial parks make simultaneous qualitative and quantitative characterization of them an unsurpassed challenge. Thus, it is of urgent need to develop advanced techniques to monitor VOCs emissions in chemical industrial parks.

Highly effective analytical techniques, e.g., gas chromatography (GC), spectrometry, and high-performance liquid chromatography (HPLC), have been used to detect VOCs.

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However, they have some drawbacks that limit their applications, such as high cost, lack of portability [4], [5]. Gas sensors based on different working principles, including metal oxide semiconductor (MOS), electrochemical (EC) and optical [6], are more suitable for rapid and online detection in practical applications in spite of their possibly lower accuracy. Because of the cross-sensitivity of gas sensors, a single gas sensor is susceptible to interference from non-target gases, while the sensor array comprised of various gas sensors with different selectivity has proven to be a powerful tool for the detection of gas analytes [7]. A sensor array system is made up of an array of chemical sensors with partial specificity and an appropriate pattern recognition system, capable of recognizing simple or complex odors [8]. Many existing sensor array systems contain gas sensors based on only one working principle for the simplicity of implementation [9]–[11]. In fact, gas sensors based on different working principles have their own advantages. MOS sensors, for example, are sensitive to most VOCs, EC sensors have better selectivity, and photo ionization detector (PID) is usually used to characterize total VOCs (TVOC). There are lower correlations between responses of a hybrid sensor array (e.g., a combination of MOS, EC, PID) than a single-type sensor array, which provides more effective features and then strengthens the power of pattern recognition, leading to better detection performance [12]–[14].

The gas sensor array technique has been widely studied and applied in medical diagnosis [15]–[17], food quality monitoring [18], [19], and environmental monitoring [20]–[23]. Many previous studies have focused on the component identification of gas mixtures [24]–[26]. However, studies on quantitative analysis of gas mixtures are fewer. Fan *et al.* used support vector regression (SVR) model optimized by particle swarm optimization (PSO) to carry out concentration estimation of methane and ethylene mixtures based on the publicly available dataset from University of California Irvine [27]. Whereas, the gas concentrations in these studies were relatively high. Although some gas sensing materials were capable of detecting ppb- even ppt-level VOCs [28], [29], the concentration estimation of low-concentration ($\leq$ 5 ppm) VOCs with gas sensor arrays has been rarely reported. For example, Zhao *et al.* [3] investigated gas mixtures with concentrations ranged from 10 ppm to 200 ppm. As for the organic vapor mixtures commonly appearing in the chemical industrial parks (e.g., multicomponent solvent), the qualitative and quantitative measurement was conducted at even higher concentration ($\sim$10000 ppm) such as in [30]. This is not enough for low-ppm cases such as fugitive emissions in the workshop or tank field which are however rather harmful to workers. As a matter of fact, the quantification of low-concentration VOCs mixtures presents great challenges and difficulties. The lower the concentrations of target gases, the smaller the differences between sensors responses [31].

In this work, a hybrid sensor array by the integration of MOS, EC, and PID sensors was built to detect low-concentration ($\leq$ 5 ppm in all cases and even $\leq$ 1 ppm in most cases) VOCs commonly found in a Chinese fine chemical industrial park, such as toluene (PhMe), dichloromethane (DCM), and ethyl acetate (EA), which are widely used as solvents and raw materials. SVM/SVR and back-propagation artificial neural network (BP-ANN) models were trained for the component identification and concentration estimation based on different input features. The remaining of this paper is organized as follows: Section II illustrates the sensor array system and data analysis method. Section III presents the results for the qualitative and quantitative analysis of different target gases. Finally, section IV reaches a conclusion.

TABLE I
SENSITIVE CHARACTERISTICS OF THE SIX SENSORS USED IN THIS STUDY

No.	Sensor model	Sensor type	Target gas	Manufacturer
1	TGS 2600	MOS	Air pollutants (e.g., C_2H_5OH)	Figaro (Japan)
2	TGS 2602	MOS	Air pollutants (e.g., VOC)	Figaro (Japan)
3	TGS 2620	MOS	VOCs	Figaro (Japan)
4	TGS 822	MOS	VOCs	Figaro (Japan)
5	4ETO	EC	C_2H_4O	CITY (UK)
6	PID-A1	PID	VOCs with ionization potentials $<$10.6 eV	Alphasense (UK)

II. MATERIAL AND METHODS

A. Gas Sensor Array

The sensor array is based on the cross-sensitivity and broad-spectrum response of gas sensors, and it is reasonable to select sensors with various response patterns. For low-concentration gas detection, sensors with high sensitivity are preferred. Therefore, after comparing the sensitive characteristics claimed by the manufacturers and obtained in our pretests, six sensors (4 MOS sensors, 1 EC sensor, and 1 PID in Table I) with high sensitivity to VOCs were selected from many commercial products to construct the hybrid sensor array. The combination of the sensors with different construction techniques are expected to improve the performance of gas classification and concentration estimation. In the MOS sensors (TGS 2600, TGS 2602, TGS 2620, TGS 822), the physicochemical reaction between target gases and sensing materials will lead to the change of the conductivity of the sensing materials. They are sensitive to a variety of VOCs with different response patterns. An EC sensor (4ETO) with three electrodes was then adopted, where the target gases undergo the redox reaction on the electrodes to form a current signal. There is also a reference electrode in the 4ETO to keep the working electrode at a fixed potential and improve the signal stability. Finally, we also added a PID (PID-A1) to construct our hybrid sensor array because it could present the TVOC concentration. Exactly, VOCs with ionization potentials below 10.6 eV are ionized to form a current in the PID. Note that the original signals of MOS sensors are conductivity changes while those of EC sensor and PID are current changes. All these signals were transferred to voltage by the signal conditioning board for the pattern recognition thereafter.

B. Preparation of Samples

Fig. 1(a) presents the monthly frequency of the occurrence of different VOCs from workshops, stationary sources, and

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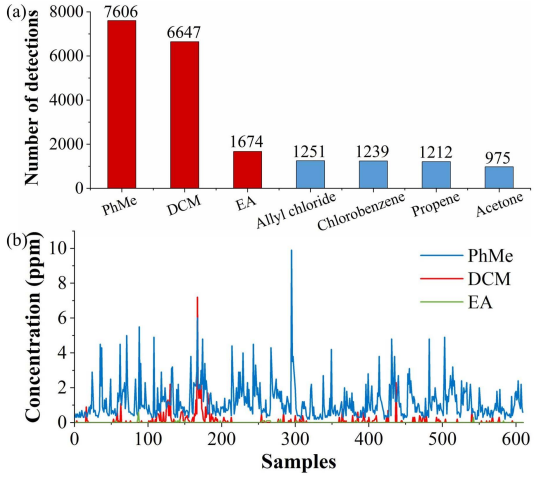


Fig. 1. VOCs emission in a Chinese fine chemical industrial park in a month: (a) total number of detections of different VOCs in the industrial park; (b) concentrations of PhMe, DCM, and EA emissions in a workshop.

tank fields in a Chinese fine chemical industrial park measured by an online mass spectrometer system (Thermo Onix Ltd., Sentinel PRO). The measurement of a VOC is considered reliable only if its concentration is recognized as over 0.1 ppm, according to the recommended detection range. It can be seen that PhMe, DCM, and EA known as common industrial solvents and raw materials were the three most frequently detected VOCs. Hence, our gas sensor array was developed to analyze PhMe, DCM, EA, and their mixtures. The concentration characteristics in a month of the three kinds of VOCs in a workshop is shown in Fig. 1(b). It can be concluded that among the 610 samples, only one had a TVOC concentration exceeding 10 ppm, and in most situations, concentrations of the three kinds of VOCs were below 5 ppm, and even below 1 ppm. Therefore, the lab testing was carried out in the concentration range of 0.5 – 5 ppm, and only three concentration levels, including 0.5, 1 and 5 ppm, were set due to the small concentration range.

Responses of the sensor array to different types of gas mixtures were tested in our study, including single gas, binary mixture, and ternary mixture. The binary gas was a mixture of PhMe and DCM, or PhMe and EA, or DCM and EA, and the ternary gas was composed of all the gas species of PhMe, DCM, and EA. We employed A/B to represent the mixture of A and B. Take PhMe/EA as an example, it means the mixture of PhMe and EA. The concentration levels of a single gas were 0.5 ppm, 1 ppm, and 5 ppm. The concentration levels of single gas in the mixtures were 0.5 ppm, 1 ppm, and 5 ppm.

C. Gas Sensing Measurement

The schematic diagram of our experimental setup is shown in Fig. 2. The gas sensor array system developed for component identification and concentration estimation, was composed of three parts: a gas distribution module, a detection module, and a signal processing and pattern recognition module. The gas distribution module, including target gases, air and a gas blender, was designed to provide target gases under various conditions for the detection module. The detection

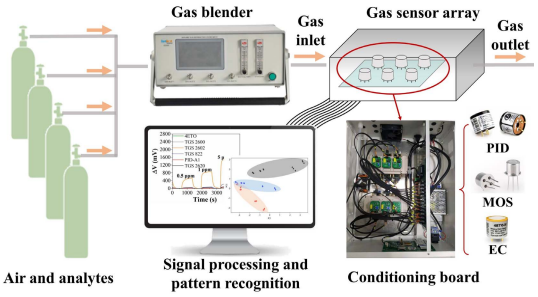


Fig. 2. Schematic diagram of the gas sensor array system.

module had a customized test chamber with gas inlet and outlet, in which a total of six gas sensors and their signal conditioning board were assembled. The signal conditioning board was adopted to transform the variations of conductivity or current into the variation of voltage through the series-connection of matching resistances to simplify the interface circuit and facilitate the data acquisition and processing. Then the signal conditioning board also ensured the output voltages were in the range of 0 – 5 V during the measurement. Sensor array voltage signals were transferred to the computer for further processing and pattern recognition.

Before the measurement, the sensors were preheated for 24 hours to make the sensor array work in a stable condition. During the tests, the gas concentrations were automatically calculated and controlled by the gas blender, and the gas flow rate was kept at constant 1200 mL/min. In a measurement, the test chamber was first cleaned by standard air with a flow rate of 1200 mL/min for 30 minutes to obtain stable sensor baseline voltages. Then, the target gases under various conditions provided by the gas blender were introduced to the sensor array for 12 minutes to obtain the sensor array responses. Finally, the sensor array was cleaned by standard air for 10 minutes to recover to its initial state.

D. Data Analysis and Statistics

Each concentration of each target gas was measured in triplicate. Therefore, 189 samples were obtained totally. The original response data were collected at a sampling frequency of about 1 Hz, and the last 5s data at the measurement step were averaged and the data then subtracted the baseline voltage in order to eliminate the influence of baseline voltage. The data were standardized using (1) and the standardized data were employed as input features to perform pattern recognition.

$$X_{ij} = \frac{x_{ij} - \mu_i}{\sigma_i} \tag{1}$$

In (1), X_{ij} is the standardized response of the sensor i to the sample j, x_{ij} is the response of the sensor i to the sample j, μ_i represents the mean of the responses of the sensor i, and σ_i is the standard deviation of the responses of the sensor i.

In multivariate statistical analysis, correlation coefficient is usually employed to demonstrate the relationship between

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different variables. The correlation coefficient is defined by

$$C_{it} = \frac{\sum_{j=1}^{J} \left(x_{ij} - \bar{x}_i\right)\left(x_{tj} - \bar{x}_t\right)}{\sqrt{\sum_{j=1}^{J} \left(x_{ij} - \bar{x}_i\right)^2} \sqrt{\sum_{j=1}^{J} \left(x_{tj} - \bar{x}_t\right)^2}} \tag{2}$$

where C_{it} is the correlation coefficient corresponding to the sensor i and sensor t, x_{ij} is the response of the sensor i to the sample j, $\bar{x}_i$ represents the average of the responses of the sensor i. When C_{it} is equal to 1, it means that the two variables are linearly correlated. We adopted correlation coefficient analysis to evaluate the relevance between sensors responses [32].

For the preliminary visualization and dimension reduction, principle component analysis (PCA) and linear discriminant analysis (LDA) were used [33], [34], while BP-ANN [35], [36] and SVM/SVR [27], [35], [37], [38] were employed for component identification and concentration estimation. The BP-ANN consisted of an input layer with six nodes or four nodes, a hidden layer with fifteen nodes and an output layer with three nodes. We adopted ReLU function as activation function. SVM was the mainstream solution of pattern recognition before neural network, so we used SVM with radial basic function (RBF) kernel to train pattern recognition model and compared its performance with BP-ANN. In addition, the effects of different features, including steady-state responses (SS), as well as the first four principal components (PCs) and the first four discriminant functions (DFs), on the performance of models were also compared and analyzed. Python 3.7 software (Python Software Foundation, USA; pytorch) was used for the data analysis and pattern recognition modelling.

III. Results and Discussion

A. Sensor Responses Analysis

Fig. 3 presents typical dynamic response curves (denoted by the voltage change, ΔV) when the sensor array was exposed to PhMe, DCM, EA in the concentration of 0.5, 1, and 5 ppm at the interval. In accordance with the overall response process, the baselines of the sensors are relatively stable and the responses increase with the gas concentrations. It is worth noting that sensors respond to different target gases in various patterns. The EC sensor, 4ETO, is sensitive to PhMe, while its voltage is almost unchanged when being exposed to DCM and EA (more clearly shown in Fig. 4). MOS sensors have different sensitivities to these three species. TGS 2600, for example, is more sensitive to DCM than PhMe and EA, however, TGS 2602 is less sensitive to DCM compared with PhMe and EA.

Fig. 4 summarizes the responses of all the six sensors to concentrations of 0.5, 1, 5 ppm PhMe, DCM and EA in a bar chart. In order to display the responses of 4ETO more clearly, the stable response values of 4ETO are magnified by a factor of 20 in the Fig. 4. It can also be seen that the sensor array has distinct voltage changes to each of the three target gases. Each concentration of each target gas was measured three times to verify the repeatability of the sensor array. It can be inferred from the error bars that the sensor array has good repeatability and the sample-to-sample variations are within

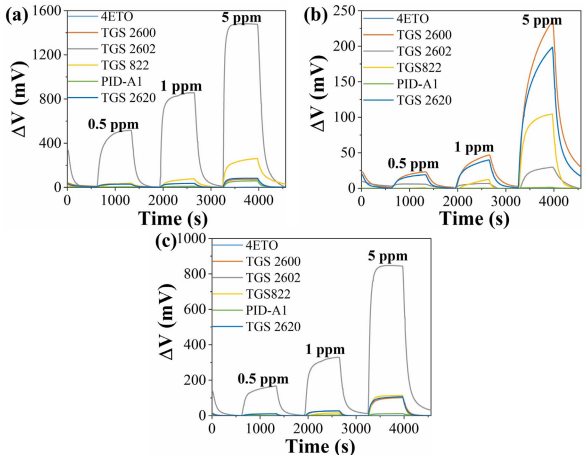


Fig. 3. Dynamic response curves of the gas sensor array: (a) PhMe, (b) DCM, (c) EA.

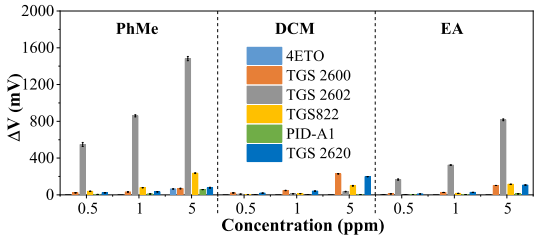


Fig. 4. Responses of the sensor array to different target gases. The responses of 4ETO are magnified by a factor of 20.

the acceptable range, since about 90% of the sample-to-sample variations among the three measurements are in the range of 0.01% – 5%, and only less than 1% of the sample-to-sample variations are greater than 10%.

The correlation coefficient analysis of sensor responses is displayed in Fig. 5. With regard to the data analysis and pattern recognition of a sensor array system, it is necessary to ensure that the correlations between sensors responses are as small as possible [12]. The correlation coefficients between EC sensor and MOS sensors are mostly smaller than those between MOS sensors, while the correlation coefficients between PID sensor and other sensors, except for TGS 2600 and TGS 2620, are all larger than 0.8. TGS 2600 is in very high correlation with TGS 2620 with a correlation coefficient up to 0.99. In general, to a certain degree, the ratio of correlation coefficients larger than 0.8 is reduced by the hybrid sensor array, which may improve the pattern recognition performance.

To sum up, the sensor array has good detection performance and cross-sensitivity characteristics for PhMe, DCM, and EA. These sensors show different responses to different target gases. Therefore, they meet the basic requirements of sensor array selection and are suitable to build a sensor array. In other words, the sensor array developed in our study is effective in distinguishing PhMe, DCM and EA.

B. PCA and LDA Feature Extraction

PCA and LDA, as two effective dimension reduction and data visualization methods, were adopted to extract features

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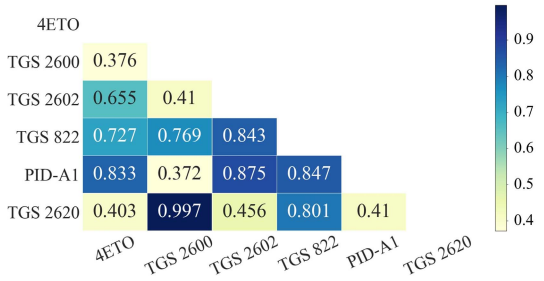


Fig. 5. Correlation coefficient analysis of sensor responses.

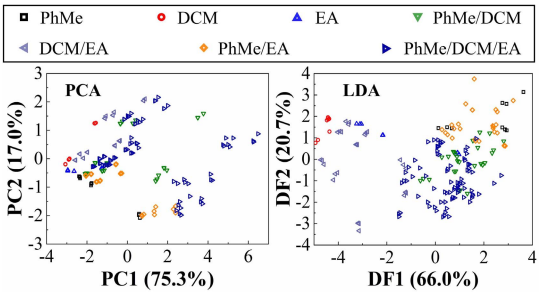


Fig. 6. PCA and LDA analysis of sensor array responses.

from the original responses. As is shown in Fig. 6, we divide our samples into seven groups, including PhMe, DCM, EA, PhMe/DCM, PhMe/EA, DCM/EA, PhMe/DCM/EA according to the kinds of VOCs contained in the target gases. PC1 and PC2 (by PCA) explain 75.3%, 17.0% of the variance of the response data, respectively, and the first two PCs account for 92.3% of the total variance of the data. And DF1 and DF2 (by LDA) explain 66.0%, 20.7% of the variance of the response data, respectively, and the first two DFs account for 86.7% of the total variance of the data.

C. Qualitative Analysis of Gas Mixture

In the identification of different VOCs and their mixtures, we compared the performances of SVM and BP-ANN with different input features, including SS, the first 4 components preprocessed by PCA or LDA. The dataset was split into training and testing subsets, and 40% data, including 76 samples, were combined into the testing set. Fig. 7 shows the confusion matrices that illustrate the gas identification accuracies with different input features and classification algorithms. The elements on the diagonal represent the percentage of samples of which the predicted labels are the same as the true labels, which are gas identification accuracies, whereas the elements off the diagonal are error rates. We adopted A-B to represent B algorithm with A as input features. For example, LDA-ANN represented BP-ANN with DFs as input features. It can be observed from Fig. 7 that all the six models show favorable classification accuracies. The accuracies of SS-ANN and SS-SVM are 93% and 92% respectively, which are the highest and second highest in the several models. Although the SS-ANN has one-third misjudgments for PhMe and EA, the overall accuracy of SS-ANN is still the highest, which may be due to the small number of single gas samples. PCA and LDA were employed for dimension reduction, and the first 4 PCs and DFs replaced SS as input features. PCA-ANN has the best performance with an accuracy of 91%, which is only 2% lower than that of SS-ANN, while the accuracy of LDA-ANN is the lowest (86%). Besides, accuracies of some substances are lower than those of others. For PhMe, DCM, and EA, the majority of the six models exhibit excellent performance with accuracies of 100%, while for PhMe/EA, all the six models have misjudgments and about 10% of the PhMe/EA samples are mistaken by SS-ANN and LDA-ANN which have the highest accuracies for PhMe/EA.

TABLE II
PERFORMANCE OF COMPONENT IDENTIFICATION OF DIFFERENT PATTERN RECOGNITION MODELS

	SS		PCA		LDA	
	SVM	BP-ANN	SVM	BP-ANN	SVM	BP-ANN
Validation set	94.2%	93.7%	94.2%	91.3%	91.2%	84.3%
Training set	100%	100%	100%	99.1%	100%	98.2%
Testing set	92.1%	93.4%	89.5%	90.8%	90.8%	85.6%

A 10-fold cross validation was carried out to further evaluate the gas identification accuracies with proposed methods. Table II compares the performances of different models. Similar to the results in Fig. 7, the performances of models with the first 4 PCs are quite close to those with SS, which are enough to meet the needs of classifying different samples. PCA-SVM and SS-SVM show the best performance with validation accuracies of 94.2%, and the accuracy of PCA-ANN is 91.3%, only 3% lower than those of the best models. In general, the pattern recognition models and sensor array developed in our study are effective in classifying different kinds of gas samples, and the classification performance of the first 4 components preprocessed by PCA are close to that of the original SS, while the performance of the first 4 DFs are poorer compared with that of SS.

D. Quantitative Analysis of Gas Mixture

The concentration estimation of target gases was carried out after component identification, using SVR and BP-ANN with different input features, including SS, the first 4 PCs, and the first 4 DFs. The dataset was split into training set (60%) and testing set (40%). Since the 76 testing samples were randomly obtained from all the 189 samples, they could be regarded as 76 independent measurements after the pattern recognition model was trained. Thus, these measurements were suitable to verify the effectiveness of the proposed sensor array to quantify the target VOCs. Note testing samples may be single, binary, and ternary gases.

Fig. 8 illustrates the predicted results of the six models employed for the concentration estimation of target gases. The predicted concentrations are close to actual ones with input features of SS and the first 4 PCs, at all the concentration levels of 0, 0.5, 1, and 5 ppm (0 means the specific gas does not exist in the sample). However, the features extracted by LDA have poorer performance, which means SS and PCs but not DFs should be selected for this application, and PCA is more recommended to perform the dimension reduction.

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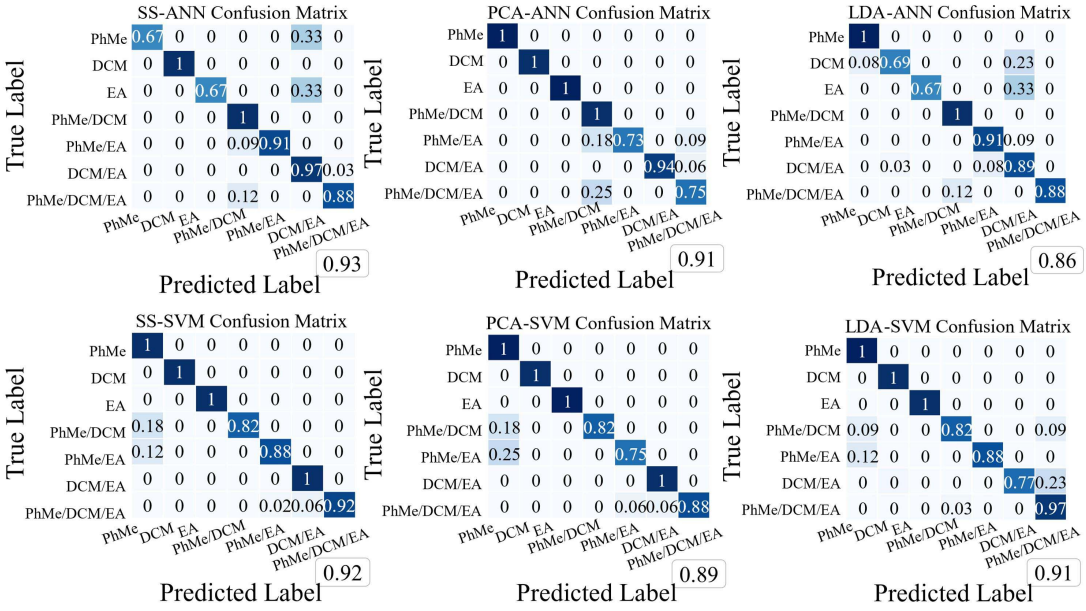


Fig. 7. Normalized Confusion matrices with six models.

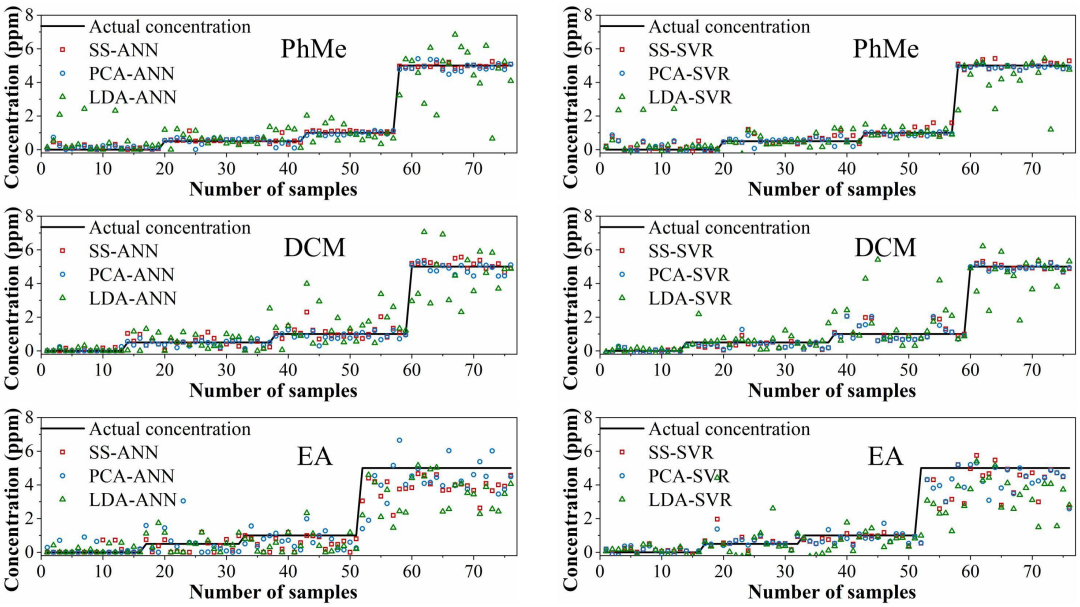


Fig. 8. Comparison of the concentration estimation performances of PhMe, DCM, and EA with different input features and different models.

As for the regression models based on BP-ANN and SVR, they both perform well for the quantitative analysis, which indicates the proposed hybrid sensor array combined with proper pattern recognition algorithm is competent to measure low-concentration VOCs mixtures, as is presented in Fig. 1(b) with concentrations of many cases even below 1 ppm (rare in previous investigations). From the perspective of target gases, PhMe and DCM are better characterized than EA with all the six models.

In order to avoid the possible bias caused by random division of training and testing set, a 10-fold cross validation was also conducted to further compare the concentration estimation performance of different input features and different models. Then the determination coefficient (R^2) and root mean squared error (RMSE) are calculated to evaluate the model performance:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \tag{3}$$

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m} (y_i - \hat{y}_i)^2} \tag{4}$$

In above equations, $\hat{y}_i$ is the estimated concentration, $\bar{y}$ is the mean of the target concentration, y_i is the target

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TABLE III
PERFORMANCE OF CONCENTRATION ESTIMATION OF DIFFERENT PATTERN RECOGNITION MODELS

		SS		PCA		LDA	
		SVR	BP-ANN	SVR	BP-ANN	SVR	BP-ANN
Validation set	R^2	**0.9544**	**0.9532**	0.9395	0.9341	0.8235	0.8015
	RMSE (ppm)	**0.4245**	**0.4193**	0.4891	0.5068	0.8369	0.8793
Training set	R^2	0.9580	0.9926	0.9489	0.9576	0.9395	0.9530
	RMSE (ppm)	0.3833	0.1641	0.4222	0.3836	0.4661	0.4207
Testing set	R^2	0.9235	0.9400	0.9395	0.9308	0.6968	0.7296
	RMSE (ppm)	0.5769	0.5116	0.5121	0.5538	1.1112	1.0434

concentration, and m represents the total number of samples. The value of R^2 ranges from 0 to 1, and the closer it is to 1, the better the regression model performance is. On the contrary, the smaller the value of RMSE, the better the regression model performance. R^2 and RMSE of training, validation, and testing sets with all six models are summarized in Table III. Take the validation set as the measurements. LDA-ANN and LDA-SVR have the worst performance in the validation set, although the two models perform well in the training set with R^2 of about 0.95 and 0.94, respectively. The poor performance of the first 4 DFs may be related to the distribution characteristic of sample data. Then follow the models with input features of the first 4 PCs, that is PCA-SVR and PCA-ANN. Therefore, PCA is more suitable for the dimension reduction in this study. SS-SVR and SS-ANN are two best models with R^2 of about 0.95 and RMSE of about 0.42 ppm (in bold in Table III). In the two models, SS-ANN has better performance in the testing set containing 76 testing samples, with R^2 of about 0.94 and RMSE of about 0.51 ppm. When it comes to the target gases, the RMSE of PhMe, DCM and EA of the testing set are about 0.15, 0.30 and 0.82 ppm, respectively. The RMSE of EA is much greater than those of PhMe and DCM, which is also indicated in Fig. 8. The reason may be that the majority of the six sensors have distinct response patterns to PhMe and DCM, but have similar response patterns to EA (see Fig. 3). As a result, although the qualitative detection performance of EA is good (see Fig. 7), its concentration estimation performance is poor. Therefore, not only the array's response patterns to different gases but also the response patterns of different sensors to the same gas should be considered in the future. In addition, when the gas concentration is the lowest target concentration (0.5 ppm), the overall RMSE of SS-ANN is about 0.28 ppm, and the RMSE of PhMe, DCM and EA are about 0.18, 0.28 and 0.40 ppm, respectively. Although the RMSE is not very small compared to the lowest target concentration, we think it is acceptable for the low-concentration cases.

IV. CONCLUSION

Previous investigations on qualitative and quantitative characterization of gas mixtures in literatures are mainly applied to high-concentration conditions (usually $\geq$10 ppm). Similar research aiming to the low-concentration cases below 5 ppm (even below 1 ppm) is rare, which is however desired for environment monitoring in chemical industrial parks. In this study, a hybrid sensor array with six gas sensors based on different working principles (MOS, EC and PID) was developed for component identification and concentration estimation of low-concentration VOCs of PhMe, DCM, and EA. BP-ANN and SVM/SVR with different input features, including SS, the first 4 PCs and DFs, were employed to implement qualitative and quantitative analysis of multiple kinds of gases. A 10-fold cross validation was adopted to statistically evaluate the performance of the models. Generally speaking, SS-SVM/SVR and SS-ANN present the best performance for both qualitative and quantitative analysis, and the identification accuracy and R^2 are about 94% and 0.95 for the validation set, respectively. The error of concentration estimation (denoted by RMSE) could be as small as 0.42 ppm. Besides, PCA is more suitable for dimension reduction than LDA in this study. The performance of models with the first 4 PCs are close to that of models with SS.

Some optimizations are yet to be considered in the future work. For example, more considerations such as the influence of temperature, moisture, and other disturbing gases may be taken into account. Also, more sensors with array optimization can be used to improve the performance and measure more VOCs species.

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